

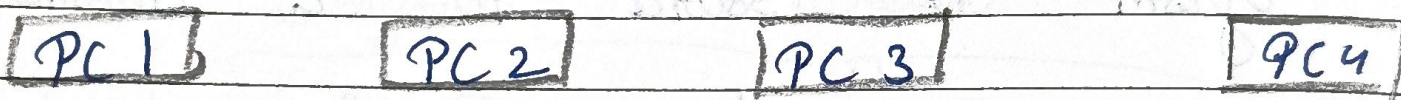
Assignment : 2 Computer network

1. Explain the Concept of Network Topology. Describe different type of network topologies such as Bus, Star, Ring, Mesh and Tree with suitable diagram.

Ans: Network topology refers to the physical or logical arrangement of Computer, Cables, and network devices in a network.

Types of Network Topologies:

1. Bus Topology:



Backbone Cable

- i. All device connect to a single backbone cable.
- ii. Data travels in both direction in the cable.
- iii. Terminators are placed at both ends.
- iv. Simple and easy to install
- v. Requires less cable than other topologies
- vi. Failure of backbone stops the entire network used in small networks.

2. Star Topology:

PC1

PC2

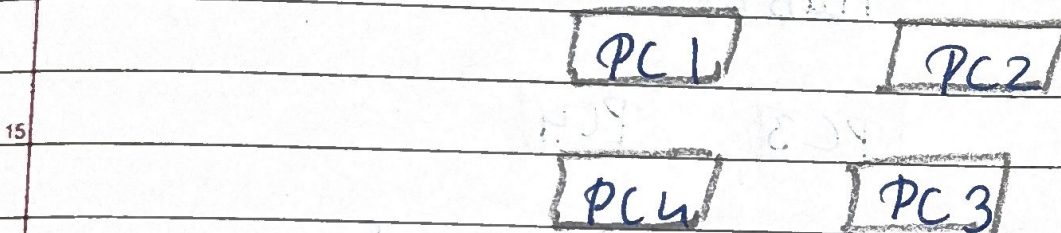
Switch / Hub

PC3

PC4

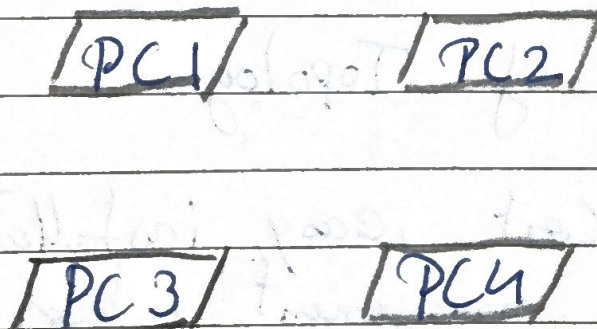
- i. All device Connect to a Central hub or Switch.
- ii. Data pass through the Central device.
- iii. Easy to install and manage.
- iv. Failure of one node does not affect other.
- v. Central device failure stops the network.
- vi. Easy to add or remove device.
- vii. Most Commonly used topology today.

3. Ring Topology.



- i. Each device Connects to two neighbouring device
- ii. Forms a closed loop (ring)
- iii. Data moves in one direction around the ring
- iv. uses token passing method
- v. Equal access to all devices.
- vi. Difficult to troubleshoot.

Mesh Topology:

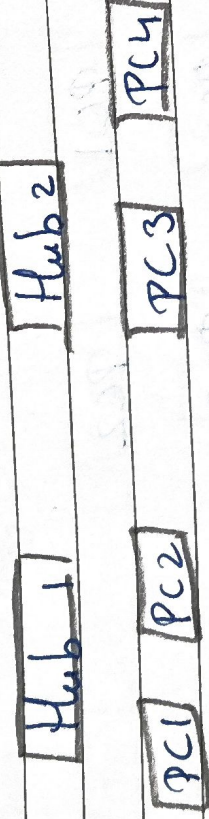


- i. Each device Connects to every other device.

- ii. provides multiple paths for data transmission.
 - iii. Very reliable and fault tolerant.
 - iv. High data security another path is used.
 - v. If one link fails, no. of cables. Requires large no. of cables to implement.
- Very expensive to implement.

5. Tree Topology.

Main Hub /



1. Combination of Star and Bus topology.
 2. Device are arranged in hierarchical structure.
 3. Multiple star networks connected to backbone.
 4. Easy network expansion.
 5. Suitable for large organization.
 6. Backbone failure affects entire network.
- Complex configuration.

2. Advantages and Disadvantages of Topologies.

1. Advantages of Topologies

- i. Bus: low cost, easy installation, less cable.
- ii. Star: Easy management, high performance, easy expansion.

- iii. Ring: Equal arcs, orderly data transfer
- iv. Mesh: Very reliable, high fault tolerance, secure

2. Disadvantages of Topologies

- i. Bus: Backbone failure stops network, difficult troubleshooting
- ii. Star: Central hub failure stops network, higher cost.
- iii. Ring: one node failure affects network, difficult to maintain.
- iv. Mesh: very expensive, complex installation.

Create a hybrid network topology model for an organization that combines star and bus topologies. Illustrate how the hybrid structure improves network efficiency and fault tolerance.

PC

PC2

Switch 1

(Bus backbone)

Switch 2

PC3

PC4

A hybrid topology combines two or more network topologies.

Here, Star networks are connected using a

Bus back bone.

+ Advantages:

1. Combines benefits of multiple topologies.
2. Improves network flexibility.
3. Easy network expansion.
4. Failure in one star network does not affect other.
5. Better fault tolerance.
6. Supports large organization.
7. Improves network efficiency and performance.

2.

- Comparison:

1. Bus:

Cost: Low
Scalability: Limited
Reliability: Low
Performance: Low when traffic increases

2. Star:

Cost: Medium
Scalability: High
Reliability: High
Performance: High

3. Ring:

Cost: Medium
Scalability: Moderate
Reliability: Medium
Performance: Moderate

4. Mesh:

Cost: very high

Scalability: low

Reliability: very high

Performance: very high